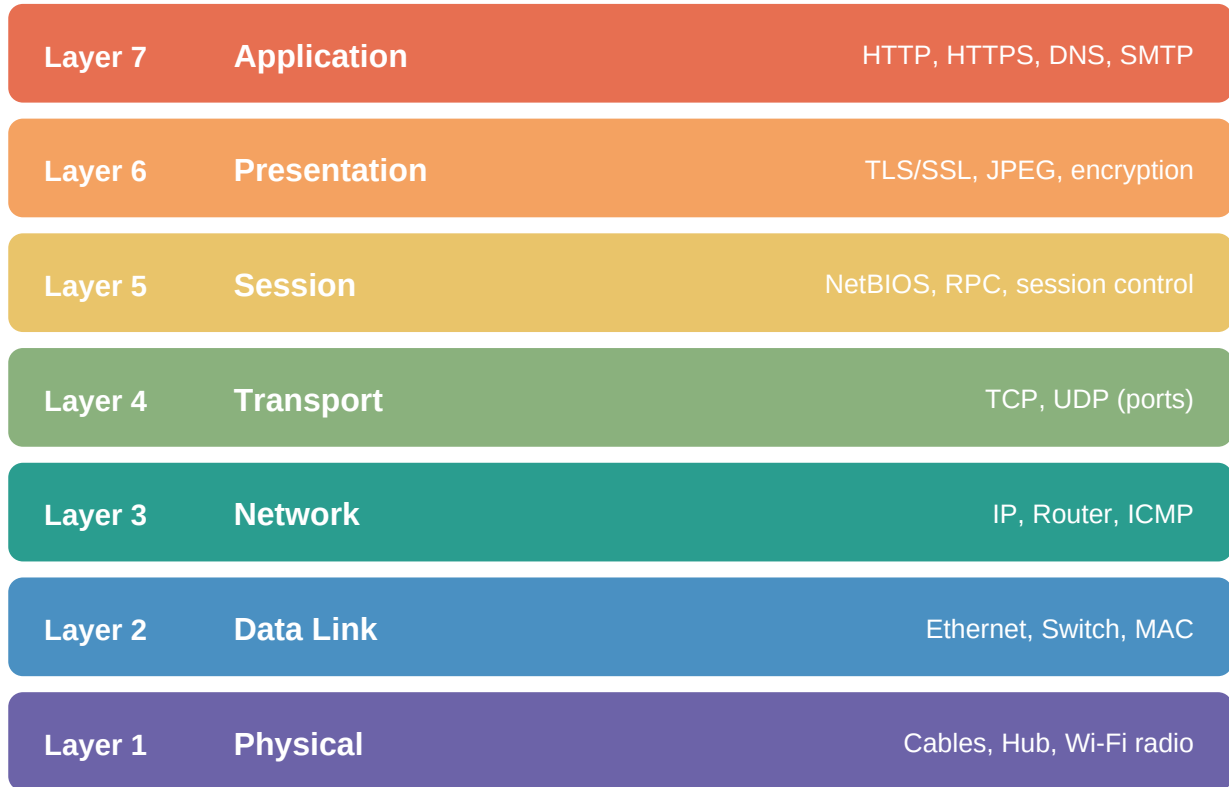


OSI Layer: Answers

1. OSI 7-Layer Model Diagram

OSI 7-Layer Model



Tip: 'All People Seem To Need Data Processing' = Application to Physical

Layer	Name	Device / Protocol example
7	Application	HTTP (web browsing), DNS, SMTP (email)
6	Presentation	TLS/SSL encryption, JPEG/MP4 formatting
5	Session	RPC / NetBIOS - opens, maintains and closes sessions
4	Transport	TCP (reliable) and UDP (fast) with port numbers
3	Network	Router, IP addressing, ICMP (ping)
2	Data Link	Switch, Ethernet, MAC addresses, Wi-Fi (802.11)
1	Physical	Ethernet cables, Hub, fibre optic, Wi-Fi radio signals

2. Scenario Matching

Scenario	Answer
a) Encrypting data before sending	Presentation Layer (6)
b) Routing packets across networks	Network Layer (3)
c) Converting data to electrical signals	Physical Layer (1)
d) Opening a web page in your browser	Application Layer (7)
e) Detecting and correcting errors in frames	Data Link Layer (2)

Format: a) Encrypting data: Presentation (6) | b) Routing packets: Network (3) | c) Electrical signals: Physical (1) | d) Opening a web page: Application (7) | e) Error detection in frames: Data Link (2)

3. WhatsApp and the OSI Layers

When you send a message on WhatsApp, the data travels **down** the layers on your phone and **up** the layers on your friend's phone:

Layer	What happens in WhatsApp
Application (7)	The WhatsApp app creates your message and uses its messaging protocol to talk to WhatsApp servers.
Presentation (6)	The message is end-to-end encrypted; photos and videos are compressed and formatted (JPEG, MP4).
Session (5)	A logged-in connection (session) with the server is established and kept alive so messages arrive instantly.
Transport (4)	TCP splits the data into segments, adds port numbers (e.g. 443) and makes sure every piece arrives in order.
Network (3)	IP adds your phone's and the server's IP addresses; routers forward the packets across the internet.
Data Link (2)	Packets are put into frames with MAC addresses and sent over Wi-Fi or mobile network to the next device.
Physical (1)	Bits are converted into radio waves (Wi-Fi/4G/5G) or light/electrical signals in cables.

Summary: Application, Presentation and Transport handle *what* is sent and *how reliably*; Network, Data Link and Physical handle *where* and *how* it physically travels.

4. Website Not Opening, but Ping to Its IP Works

Observation: A successful ping (ICMP) means the connection works through the **Physical (1), Data Link (2) and Network (3)** layers: cables/Wi-Fi are fine, frames are delivered, and IP routing works. So the problem is in the **upper layers (4 to 7)**.

Likely layer	Reasoning
Application (7) - most likely	DNS failure: the browser cannot convert the domain name to an IP address (ping by IP skips DNS). Also possible: a web server/HTTP error or a wrong proxy setting.
Transport (4)	A firewall may block TCP port 80/443, so ICMP ping passes but the web connection is refused or times out.
Presentation / Session (6, 5)	A TLS/SSL certificate error or a failed handshake can stop the HTTPS page from loading.

How to check:

- Run `nslookup website.com` - if it fails, DNS (Layer 7) is the issue; try changing DNS to 8.8.8.8.
- Run `telnet website.com 443` or `curl -v` - if it fails, check the firewall/ports (Layer 4).
- Check the browser's certificate/security warning (Layers 5-6).

Conclusion: Layers 1-3 are working; the fault is in Layers 4-7, most commonly DNS at the Application layer.